Notes: van Binsbergen (JFE, 2026)

Duration-based stock valuation: Reassessing stock market performance and volatility

Contents

1 Big picture

Standard practice defines an "equity premium" as the difference between stock returns and the return on a very short-duration instrument (e.g., 1–3 month T-bills). But the cash flows of equities are *long-duration*: the present value of dividends far in the future is a large share of the stock-market value. Hence, comparing stock returns to short-term bills conflates:

- compensation for **dividend risk** (equity cash-flow risk and its pricing), and
- compensation for **term risk** (holding long-duration fixed income instead of a short bill).

The paper proposes a duration-matched benchmark: replace each risky dividend strip with a maturity-matched government zero-coupon bond, and compare realized returns and volatility. This shifts the benchmark from "cash" to "a long bond portfolio with the same duration profile as equity dividends."

2 Model objects and duration-matching logic

2.1 Dividend strips: valuation and weights

Let D_{t+n} denote the aggregate dividend paid in year $t+n$. Let $P_{t,n}$ be the time-t present value of the *expected* dividend at maturity n:

$$P_{t,n} = \frac{\mathbb{E}_t[D_{t+n}]}{\exp\big(n(y_{t,n} + \theta_{t,n})\big)}, \qquad n \geq 1. \tag{1}$$

Here $y_{t,n}$ is the continuously compounded real or nominal risk-free spot rate for maturity n, and $\theta_{t,n}$ is the (per-unit-time) dividend-risk term premium for maturity-n dividends.

The aggregate stock index value equals the sum of all dividend strips:

$$S_t = \sum_{n=1}^{\infty} P_{t,n}. \tag{2}$$

Define the strip value weights

$$w_{t,n} = \frac{P_{t,n}}{S_t}, \qquad \sum_{n=1}^{\infty} w_{t,n} = 1. \tag{3}$$

2.2 One-period returns on dividend strips and on the stock index

For a strip with maturity $n > 1$, the one-period return is

$$R^d_{t+1,n} = \frac{P_{t+1,n-1}}{P_{t,n}} - 1, \qquad n > 1. \tag{4}$$

For the $n = 1$ strip, it pays the realized dividend next period:

$$R^d_{t+1,1} = \frac{D_{t+1}}{P_{t,1}} - 1. \tag{5}$$

The one-period stock-index return is

$$R^s_{t+1} = \frac{S_{t+1} + D_{t+1}}{S_t} - 1. \tag{9}$$

Expected index return as a strip-weighted average. Because the index is (conceptually) a portfolio holding one unit of each strip,

$$\mu_t^s \equiv \mathbb{E}_t[R_{t+1}^s] = \sum_{n=1}^{\infty} w_{t,n}\, \mathbb{E}_t\left[R_{t+1,n}^d\right]. \tag{10}$$

2.3 Bond counterfactuals: matching maturity and duration

Let the price of an n-period zero-coupon bond at time t be $\exp(-ny_{t,n})$. Its one-period holding return is

$$R_{t+1,n}^b = \frac{\exp(-(n-1)y_{t+1,n-1})}{\exp(-ny_{t,n})} - 1, \tag{6}$$

with conditional expected return

$$\mu_{t,n}^b \equiv \mathbb{E}_t\left[R_{t+1,n}^b\right]. \tag{7}$$

Define the **dividend risk premium (one-period expected excess return) at maturity** n as

$$\psi_{t,n} \equiv \mathbb{E}_t\left[R_{t+1,n}^d\right] - \mu_{t,n}^b. \tag{8}$$

2.4 Main object: the long-duration dividend premium

The paper's main quantity is the unconditional expected return earned by the index-implied portfolio of risky dividends *in excess of* maturity-matched government bonds:

$$\begin{aligned}\Psi_0 &= \mathbb{E}\left[\mathbb{E}_t \sum_{n=1}^{\infty} w_{t,n}\psi_{t,n}\right] \\ &= \mathbb{E}\left[\mu_t^s - \mathbb{E}_t \sum_{n=1}^{\infty} w_{t,n}\mu_{t,n}^b\right].\end{aligned} \tag{11}$$

Sample estimator (with maturity cutoff). With an N-year maturity cutoff (because very long bond yields are imprecisely measured),

$$\widehat{\Psi}_0 \approx \frac{1}{T}\sum_{t=1}^{T}\left[\frac{S_{t+1}+D_{t+1}}{S_t} - \sum_{n=1}^{N} w_{t,n}R_{t+1,n}^b\right]. \tag{12}$$

2.5 Macaulay duration of the stock market

Given weights $\{w_{t,m}\}$, the Macaulay duration is

$$Dur_t = \sum_{m=1}^{\infty} w_{t,m}\, m. \tag{13}$$

Duration is an *accounting weight* used to build counterfactual bond portfolios (rather than an interest-rate hedging statistic for stocks).

3 Derivations under Gordon growth (and how weights are constructed)

3.1 Continuous-time Gordon growth: strip prices, weights, and duration

Assume a deterministic exponential dividend path (normalized):

$$D_t = \exp(gt), \tag{14}$$

and a constant continuously compounded expected return on the index μ_s. Then the index value equals the discounted value of the dividend stream:

$$\begin{aligned} S_t &= D_t \int_0^\infty \exp((g - \mu_s)m)\ \mathrm{d}m \\ &= \frac{D_t}{\mu_s - g}, \qquad (\mu_s > g). \end{aligned} \tag{15}$$

Equivalently, the dividend yield is

$$\frac{D_t}{S_t} = \mu_s - g. \tag{16}$$

Under these assumptions, the m-maturity dividend strip value is

$$P_{t,m} = D_t \exp\big(m(g - \mu_s)\big), \tag{17}$$

implying a weighting scheme that is independent of time t and only dependent on maturity m:

$$w_{t,m} = \frac{P_{t,m}}{S_t} = \frac{D_t \exp((g - \mu_s)m)}{S_t}. \tag{18}$$

Substituting the Gordon-growth index value in Eq. (15) yields the familiar exponential form:

$$w_{t,m} = (\mu_s - g) \exp((g - \mu_s)m). \tag{19}$$

Check that weights integrate to one. Let $a \equiv \mu_s - g > 0$. Then $w(m) = ae^{-am}$, and

$$\int_0^\infty w(m)\ \mathrm{d}m = \int_0^\infty ae^{-am}\ \mathrm{d}m = \left[-e^{-am}\right]_0^\infty = 1.$$

Duration formula. Using $w(m) = ae^{-am}$,

$$\begin{aligned} Dur &= \int_0^\infty m\, w(m)\ \mathrm{d}m = \int_0^\infty m\, ae^{-am}\ \mathrm{d}m \\ &= \frac{1}{a} = \frac{1}{\mu_s - g}. \end{aligned} \tag{20}$$

This is the paper's key heuristic: **low dividend yields imply high equity duration**.

3.2 Discrete annual weights and the cutoff mass

Empirically, bond data are observed at annual maturities. Convert $w(m)$ to annual weights by integrating over $m \in [n, n+1)$:

$$\int_n^{n+1} w(m)\ \mathrm{d}m = \int_n^{n+1} (\mu_s - g) \exp((g - \mu_s)m)\ \mathrm{d}m \tag{21}$$

$$= \exp((g - \mu_s)n) - \exp((g - \mu_s)(n+1)). \tag{22}$$

Because bond yields beyond very long maturities are noisy, impose a cutoff $CO \in \{30, 40\}$ years. The cutoff mass is defined by

$$w_t^{CO} \equiv \sum_{n=CO}^{\infty} w_{t,n}. \tag{23}$$

Operationally, the portfolio assigns this cumulative tail weight to the cutoff-maturity bond.

3.3 Dividend strip data: converting futures to spot prices and extending the curve

Dividend strips are traded as dividend futures. Let $F_{t,n}$ be the futures price (delivery at $t+n$). Convert futures prices to spot prices via the no-arbitrage relationship

$$P_{t,n} = \exp(-n y_{t,n})\, F_{t,n}. \tag{24}$$

Then weights are directly observed for traded maturities:

$$w_{t,n} = \frac{P_{t,n}}{S_t}. \tag{25}$$

The cumulative observed weight up to maturity N is

$$\sum_{n=1}^{N} w_{t,n} = \frac{P_{t,1} + \cdots + P_{t,N}}{S_t}. \tag{26}$$

Deriving the implied $(\mu - g)$ from observed cumulative weights. Under exponential weights, the continuous-time cumulative mass up to N years is

$$\int_0^N a e^{-am}\ \mathrm{d}m = 1 - e^{-aN}.$$

Setting $1 - e^{-aN}$ equal to the observed cumulative weight $\sum_{n=1}^{N} w_{t,n}$ and solving yields

$$\mu - g = -\frac{1}{N} \ln\left(1 - \sum_{n=1}^{N} w_{t,n}\right). \tag{27}$$

This pins down a terminal-value Gordon tail consistent with the observed near-term strip prices.

For $n > N$, the annual weights follow the Gordon-growth tail:

$$w_{t,n} = \exp\big(-(\mu - g)n\big) - \exp\big(-(\mu - g)(n+1)\big), \qquad n > N. \tag{28}$$

Table 1
Annual returns on nominal constant maturity zero coupon bonds in the U.S. The second row in the table lists the average annual bond returns ($\hat{\mu}_n^b$) on constant maturity zero coupon bond strategies for maturities ranging between 1 year and 30 years using annual data between 1996 and 2021. The numbers are reported in percent per year. The third row reports the annual standard deviation, also in percent. The last row reports the average annual log returns. The last column lists the corresponding statistics for the S&P500 index.

Maturity (duration) in years	1	2	5	7	10	15	20	25	30	S&P500
Mean (%)	2.44	2.98	4.74	5.69	6.79	8.16	9.39	10.82	12.65	11.80
St. Dev. (%)	2.17	2.60	4.84	6.56	9.07	12.99	17.14	22.04	27.96	17.69
Mean log (%)	2.39	2.91	4.53	5.35	6.22	7.14	7.76	8.31	8.86	9.75

4 Empirical construction: constant-maturity bonds and strip-replicating portfolios

4.1 Constant-maturity zero-coupon bonds

Given a yield curve $\{y_{t,n}\}$, compute one-period bond returns using Eq. (6). For each maturity n, the paper reports:

- mean annual return $\widehat{\mu}_n^b$,
- annual return standard deviation,
- mean annual log return,
- Sharpe ratios (using the 1-year bond as a short-duration benchmark).

4.2 Strip-replicating bond portfolios (duration-matched counterfactuals)

A counterfactual bond portfolio return is a strip-weighted average of bond returns:

$$R^b_{t+1,CF} = \sum_{n=1}^{N} w_{t,n}\, R^b_{t+1,n},$$

where weights are generated using:

- constant Gordon weights for $\mu_s - g \in \{0.02, 0.03, 0.06\}$ and $CO \in \{30, 40\}$,
- time-varying Gordon weights using the observed dividend yield D_t/S_t,
- dividend-strip-implied weights for traded maturities, plus Gordon tails (Eqs. (27)–(28)).

The realized long-duration dividend premium is then

$$\widehat{\Psi}_0 \approx \overline{R^s_{t+1} - R^b_{t+1,CF}},$$

as in Eq. (12).

5 US evidence: results and interpretation (Tables 1–5, Figs. 6–11)

5.1 Table 1: US bond return moments across maturities (1996–2021)

What Table 1 shows. Constant-maturity nominal zero-coupon bond returns for maturities 1–30 years, next to the S&P 500.

Key pattern. Mean bond returns rise with maturity, approaching (and at 30 years exceeding) the S&P 500 mean return. Bond volatility also rises sharply with maturity. Around 20 years maturity, bond volatility is already near equity volatility.

Reading. If equity duration is long (well above 10 years), a large fraction of the standard "stock minus T-bill" premium is plausibly a *term premium* comparison rather than a *pure dividend-risk* premium.

Table 2
Annual return differences between the S&P500 and U.S. constant maturity zero coupon nominal bonds. The second row in the table lists the difference between the annual returns on the S&P500 index and the annual returns on constant maturity zero coupon bonds ($\hat{\mu}_n^b$) for maturities ranging between 15 and 30 years using annual data between 1996 and 2021 in percent. The third row reports the t-statistic on the difference. All numbers are reported in percent.

Duration in years	15	20	25	30
$\hat{\mu}^s - \hat{\mu}^b$ (% Annual)	3.64	2.41	0.98	−0.85
t-stat on difference	0.71	0.42	0.15	−0.11

5.2 Table 2: the duration-matched premium (US, 1996–2021)

What Table 2 does. Computes $\widehat{\mu}_s - \widehat{\mu}_n^b$ for durations $n \in \{15, 20, 25, 30\}$, with t-statistics.

Main implication. Differences shrink rapidly with maturity; they are small and statistically weak. At 30 years, the difference is slightly negative (stocks underperform 30-year constant-maturity bonds).

Interpretation. A duration-matched notion of "equity premium" is much smaller than the conventional stock–bill premium, at least over 1996–2021.

5.3 Figure 6 and Figure 7: mean and volatility term structures (1996–2021)

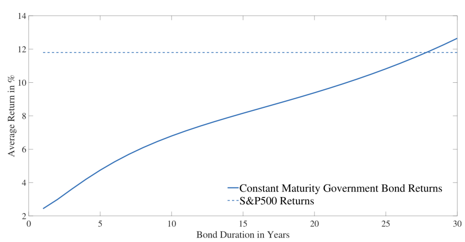

Fig. 6. Comparing annual average returns for stocks and bonds: 1996–2021. The graph plots annual average bond returns (solid line) and stock returns (dashed line) over the sample period 1996–2021.

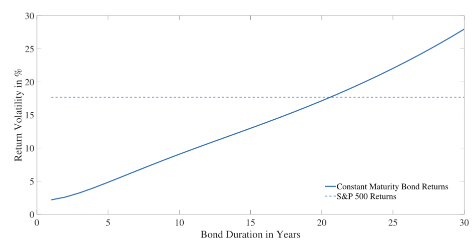

Fig. 7. Comparing annual return volatilities for stocks and bonds: 1996–2021. The graph plots the annual volatility of bond returns (solid line) and stock returns (dashed line) over the sample period 1996–2021.

- **Fig. 6** visualizes the upward slope of mean bond returns with maturity relative to the stock mean.
- **Fig. 7** shows bond volatility increasing with maturity and reaching (or exceeding) equity volatility at long maturities.

5.4 Table 3 and Figures 8–10: robustness across longer samples

Table 3 repeats mean/volatility/Sharpe comparisons for: 1996–2021, 1986–2021, and 1972–2021.

Core. Across samples, long-maturity bonds have mean returns comparable to equities, and their volatility can be similar to equity volatility at sufficiently long maturities.

Figs. 8–10 summarize these patterns:

- mean-return term structures across samples,

Table 3
Annual return moments for constant maturity nominal zero coupon bonds vs. the S&P 500 index for three samples. The second through fourth rows in the table list the average annual bond returns ($\hat{\mu}_n^b$) on constant maturity zero coupon bond strategies for maturities ranging between 1 year and 30 years using annual data for 3 different samples. The fifth through eight rows list the annual standard deviation. The last three rows report the annual Sharpe ratios using the 1-year bond as the risk free asset. The last column lists the corresponding statistics for the S&P500 index. Numbers in italics use extrapolated data implied by the NSS yield curves.

Maturity in years	1	2	5	7	10	15	20	25	30	S&P 500
Mean 1996–2021 (Annual %)	2.44	2.98	4.74	5.69	6.79	8.16	9.39	10.82	12.65	11.80
Mean 1986–2021 (Annual %)	3.59	4.27	6.21	7.26	8.55	10.31	12.00	13.89	16.15	12.85
Mean 1972–2021 (Annual %)	5.06	5.57	6.96	7.67	8.52	*9.72*	*10.84*	*11.95*	*13.39*	12.55
Vol 1996–2021 (Annual %)	2.17	2.60	4.84	6.56	9.07	12.99	17.14	22.04	27.96	17.69
Vol 1986–2021 (Annual %)	2.81	3.43	6.09	7.95	10.63	14.83	19.11	23.89	29.61	16.61
Vol 1972–2021 (Annual %)	3.74	4.25	7.03	9.17	12.50	*17.99*	*22.62*	*26.24*	*30.79*	17.18
Sharpe ratio 1996–2021	NA	0.211	0.477	0.496	0.480	0.440	0.405	0.380	0.365	0.529
Sharpe ratio 1986–2021	NA	0.198	0.430	0.462	0.467	0.454	0.440	0.431	0.425	0.558
Sharpe ratio 1972–2021	NA	0.118	0.270	0.284	0.277	*0.259*	*0.255*	*0.263*	*0.271*	0.436

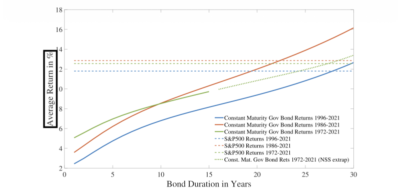

Fig. 8. Comparing annual average returns for stocks and bonds over three samples. The graph plots annual average bond returns (solid lines) and stock returns (dashed lines) over three different samples: 1972–2021 (in green), 1986–2021 (in red) and 1996–2021 (in blue). The dotted line in green represents average bond returns where the 1972–1985 bond returns are the result from NSS yield estimates that are extrapolated relative to the available bond data that GSW used to estimate the NSS curves. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

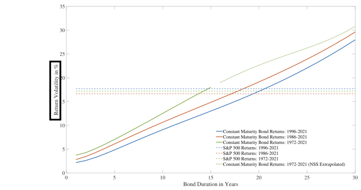

Fig. 9. Comparing annual return volatilities for stocks and bonds over three samples. The graph plots the annual volatilities of bond returns (solid lines) and stock returns (dashed lines) over three different samples: 1972–2021 (in green), 1986–2021 (in red) and 1996–2021 (in blue). The dotted line in green represents the volatility of bond returns where the 1972–1985 annual bond returns are the result from NSS yield estimates that are extrapolated relative to the available bond data that GSW used to estimate the NSS curves. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

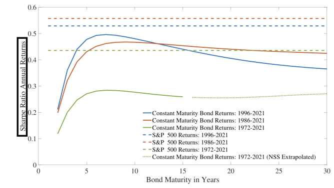

Fig. 10. Comparing annual sharpe ratios for stocks and bonds over three samples. The graph plots annual Sharpe ratios for bond returns (solid lines) and stock returns (dashed lines) over three different samples: 1972–2021 (in green), 1986–2021 (in red) and 1996–2021 (in blue). The dotted line in green represents Sharpe ratios for bond returns where the 1972–1985 bond returns are the result from NSS yield estimates that are extrapolated relative to the available bond data that GSW used to estimate the NSS curves. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 4

Strip-replicating nominal bond portfolios. The table reports the average annual returns on the strip-replicating bond portfolios using a variety of different weighting schemes using annual data between 1972 and 2021. Columns 2 through 7 use a constant Gordon growth model to generate the weights, whereas columns 8 and 9 use the real-time dividend yield on the S&P500 as the value for $\mu^s - g$ to construct Gordon growth model weights. The last two columns use dividend-strip implied weights combined with the terminal value implied Gordon growth weights. The first row reports the model inputs to the Gordon growth formula to obtain the strip weights. The second row reports the cutoff year, which either happens after 30 years or after 40 years. The third row reports the (average) duration (Dur) which is time-varying (TV) for the last four counterfactuals. The fourth row reports the average of the annual returns of the counterfactual bond portfolio, and the fifth row reports its standard deviation in percent. The second-to-last row reports the implied realized annual long-term dividend premium ($\hat{\Psi}_0$) as defined in Eq. (12) and the last row reports the difference in the annual log return between the stock index return and the counterfactual bond portfolio in percent.

Counterfact.	I	II	III	IV	V	VI	VII	VIII	IX	X
$\mu^s - g$	0.02	0.02	0.03	0.03	0.06	0.06	D_t/S_t	D_t/S_t	Strips	Strips
Cutoff year	30	40	30	40	30	40	30	40	30	40
Duration (Dur)	22.8	27.8	20.1	23.6	14.3	15.6	20.6	24.7	20.7	24.7
$\sum w_{t,n}\mu^b_{t,n}$ (% Ann.)	11.5	15.0	10.8	13.3	9.3	10.2	11.5	14.2	11.3	13.8
Std. Dev. (% Ann.)	23.4	34.2	20.7	27.5	15.3	16.7	21.0	27.0	20.7	26.4
$\hat{\Psi}_0$ (% Ann.)	1.03	−2.45	1.73	−0.71	3.23	2.38	1.04	−1.63	1.29	−1.23
Mean diff log rets	1.73	0.38	1.92	0.72	2.54	1.93	1.31	−0.16	1.51	0.12

- volatility term structures across samples,
- Sharpe ratio term structures across samples.

5.5 Table 4: strip-replicating bond portfolios (1972–2021)

Table 4 constructs 10 counterfactual portfolios (I–X) using different strip-weighting schemes and cutoffs. It reports:

- implied duration,
- mean return of the counterfactual bond portfolio,
- volatility of the counterfactual,
- realized long-duration dividend premium $\widehat{\Psi}_0$,
- mean difference in log returns between equity and the counterfactual.

Main takeaways.

- Longer-duration counterfactuals (lower $\mu_s - g$, larger cutoff) put more weight on long bonds, raising counterfactual mean returns and volatility.
- The implied $\widehat{\Psi}_0$ is often modest and can be negative for the longest-duration counterfactuals.
- Hence, realized compensation for long-duration dividend risk is not robustly large over 1972–2021.

5.6 Figure 11: cumulative relative performance

Fig. 11 plots log cumulative performance of the S&P 500 versus selected counterfactual bond portfolios. A useful reading is:

- the relative performance line is *not* a monotone "equities win" story,
- much of the realized equity premium versus bills can be replicated by moving to a long-duration bond benchmark.

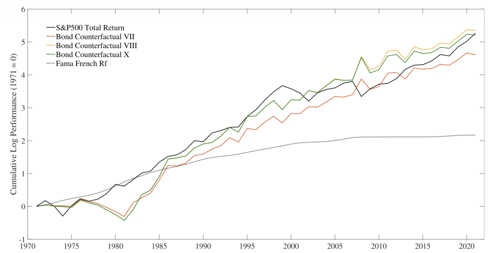

Fig. 11. Cumulative return performance of the S&P500 index relative to fixed income matched portfolio. The graphs plot the log cumulative performance between January 1972 and December 2021 of the S&P500 index and compares it to nominal fixed income counterfactuals. The fixed income counterfactuals plotted are VII, VIII and X described above, as well as the performance of the risk free rate used in the annual Fama and French factor models.

Table 5
Variance decomposition U.S. excess stock returns. The table provides a variance decomposition of U.S. stock returns in excess of the short duration fixed income instrument (in this case the risk free rate used in the Fama French factors) into the contribution of the realized term premium and the realized dividend risk premium. The long-duration fixed income counterfactual (CF) uses dividend-yield-based Gordon growth weights with a cutoff of 30 years (Counterfactual VII). In columns 2 and 4 the table reports the raw variance and covariance numbers for both the full sample as well as the 1996–2021 subsample. Columns 3 and 5 report the numbers scaled by the variance of $R^S_{t+1} - R^b_{t+1,1}$, so that the decomposition terms sum up to 100%.

	1972–2021		1996–2021	
$\mathrm{var}\left(R^S_{t+1} - R^b_{t+1,1}\right)$	0.031	100%	0.032	100%
Decomposition				
$\mathrm{var}\left(R^S_{t+1} - R^b_{t+1,CF}\right)$	0.075	244%	0.110	343%
$\mathrm{var}\left(R^b_{t+1,CF} - R^b_{t+1,1}\right)$	0.046	148%	0.043	134%
$2\mathrm{cov}\left(R^S_{t+1} - R^b_{t+1,CF}, R^b_{t+1,CF} - R^b_{t+1,1}\right)$	−0.090	−293%	−0.121	−377%

5.7 Variance decomposition and "excess volatility" (Table 5 and Eq. 29)

Using a duration-matched fixed income counterfactual CF, the paper decomposes excess stock returns relative to a short-duration instrument (here, the Fama–French risk-free rate) as:

$$R^s_{t+1} - R^b_{t+1,1} = \underbrace{(R^s_{t+1} - R^b_{t+1,CF})}_{\text{realized dividend risk premium}} + \underbrace{(R^b_{t+1,CF} - R^b_{t+1,1})}_{\text{realized term premium}}. \tag{29}$$

Taking variances implies the accounting identity:

$$\mathrm{Var}(R^s - R^b_1) = \mathrm{Var}(R^s - R^b_{CF}) + \mathrm{Var}(R^b_{CF} - R^b_1) + 2\mathrm{Cov}(R^s - R^b_{CF},\, R^b_{CF} - R^b_1).$$

Table 5 interpretation.

- Both components can be very volatile.
- The covariance term is strongly negative: the realized term premium and the realized dividend risk premium co-move negatively (correlations around −0.77 to −0.88 in the reported samples).
- This negative covariance helps explain why stock–bill excess return variance is not even larger: bond discount-rate shocks and equity "dividend premium" shocks partly offset.

6 International evidence (Tables 6–12, Figs. 12–17)

6.1 United Kingdom: nominal and real bonds (Tables 6–8)

Table 6 (UK nominal Gilts). Across 1996–2021, 1985–2021, and 1971–2021, long-maturity nominal bond returns are comparable to (and in some samples close to) equity index returns, with very high volatility at long maturities.

Table 6
Annual returns on constant maturity nominal zero coupon bonds: United Kingdom. The second row of the first panel lists the average annual bond returns ($\hat{\mu}_n^b$) on constant maturity nominal zero coupon bond strategies for maturities ranging between 1 year and 20 years using annual data between January 1996 and December 2021 in percent. The third row reports the standard deviation. The last column lists those statistics for the FTSE All Share index. The second panel repeats all statistics but now for the longer sample 1985–2021. The third panel reports the statistics for a 50-year sample 1971–2021.

Mat. in years	1	2	5	10	15	20	FTSE
Sample: 1996–2021							
Mean (% Annual)	2.84	3.29	4.83	6.92	8.65	10.10	8.03
St. Dev. (%)	2.58	2.84	4.73	8.67	12.01	14.97	15.54
Mean log (%)	2.77	3.20	4.62	6.38	7.73	8.79	6.62
Sample: 1985–2021							
Mean (%)	4.99	5.39	6.99	8.94	10.12	10.86	10.54
St. Dev. (%)	4.24	4.41	6.55	10.22	13.82	18.09	15.48
Mean log (%)	4.80	5.17	6.57	8.13	8.90	9.13	8.97
Sample: 1971–2021							
Mean (%)	6.61	7.11	8.50	10.33	12.80	17.95	14.42
St. Dev. (%)	4.77	5.53	9.66	16.47	26.02	41.96	27.59
Mean log (%)	6.30	6.74	7.79	8.83	9.58	10.57	10.86

Table 7
Inflation rate retail price index: United Kingdom. The table reports the annual mean, and volatility of the inflation rate π_t as measured by the Retail Price Index (RPI) in percent for two sample periods: 1996–2021 and 1986–2021.

	Annual mean (%)	Annual Vol. (%)
1996–2021	2.92	1.47
1985–2021	3.46	1.93

Table 8
Annual returns on constant maturity real zero coupon bonds in the United Kingdom. The second row in the top panel lists the average annual real bond returns ($\hat{\mu}_n^{rb}$) on constant maturity inflation-protected zero coupon bond strategies for maturities ranging between 1 year and 30 years using annual data between January 1996 and December 2021. The third row reports the annual standard deviation. The last column lists those statistics for RPI inflation-adjusted (π_t) FTSE index returns. The second panel repeats the same numbers for the 1986–2021 sample period.

Maturity in years	1	2	3	4	5	7	10	15	20	25	30	FTSE-π_t
Sample: 1996–2021												
Mean (%)	0.50	0.64	1.08	1.48	1.83	2.41	3.16	4.36	5.58	6.86	8.22	5.11
St. Dev. (%)	2.67	2.94	3.13	3.36	3.60	4.08	4.92	6.63	8.55	10.73	13.21	15.13
Mean log (%)	0.47	0.60	1.03	1.42	1.75	2.30	3.01	4.08	5.13	6.18	7.22	3.75
Sample: 1986–2021												
Mean (%)	1.24	1.46	1.87	2.24	2.56	3.07	3.69	4.65	5.64	6.69	7.82	6.88
St. Dev. (%)	2.63	3.04	3.32	3.60	3.88	4.44	5.33	7.09	9.10	11.36	13.89	15.37
Mean log (%)	1.20	1.40	1.80	2.16	2.45	2.93	3.50	4.33	5.13	5.93	6.74	5.38

Table 7 (UK inflation). Defines annual inflation from the Retail Price Index (RPI):

$$\pi_t = \frac{RPI_t}{RPI_{t-1}} - 1. \tag{30}$$

Table 8 (UK real bonds). Inflation-protected bonds allow a real-return comparison. The table shows that sufficiently long real bonds can have mean returns comparable to real equity returns, while real bonds still exhibit meaningful volatility due to real-rate movements.

6.2 Europe: Germany and Eurostoxx (Tables 9–10)

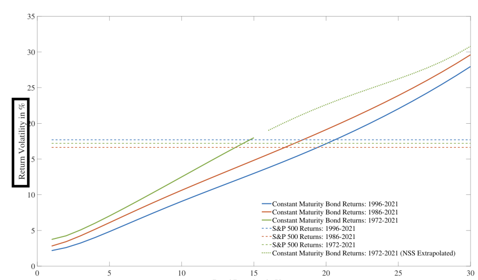

Fig. 9. Comparing annual return volatilities for stocks and bonds over three samples. The graph plots the annual volatilities of bond returns (solid lines) and stock returns (dashed lines) over three different samples: 1972–2021 (in green), 1986–2021 (in red) and 1996–2021 (in blue). The dotted line in green represents the volatility of bond returns where the 1972–1985 annual bond returns are the result from NSS yield estimates that are extrapolated relative to the available bond data that GSW used to estimate the NSS curves. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

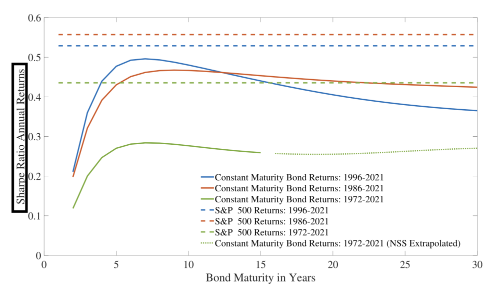

Fig. 10. Comparing annual sharpe ratios for stocks and bonds over three samples. The graph plots annual Sharpe ratios for bond returns (solid lines) and stock returns (dashed lines) over three different samples: 1972–2021 (in green), 1986–2021 (in red) and 1996–2021 (in blue). The dotted line in green represents Sharpe ratios for bond returns where the 1972–1985 bond returns are the result from NSS yield estimates that are extrapolated relative to the available bond data that GSW used to estimate the NSS curves. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 9. German constant-maturity bond returns across maturities compared to Eurostoxx. A striking feature is that even 20-year bonds can outperform Eurostoxx over 1996–2021, with volatilities that are also high.

Table 10. Directly reports Eurostoxx minus bond return differences for long maturities. The differences shrink and become negative at longer maturities (stocks underperform long bonds).

6.3 Japan: bonds vs Topix (Tables 11–12)

Table 11. Japanese bond returns are low in level for short maturities in the recent sample, but long maturities still earn nontrivial returns. Equity volatility (Topix) is extremely high relative to bond volatility.

Table 11
Annual returns on constant maturity zero coupon bonds: Japan. The second row of the first panel lists the average annual bond returns ($\hat{\mu}_n^b$) on constant maturity zero coupon bond strategies for maturities ranging between 1 year and 15 years using annual data between January 1996 and December 2021. The third row reports the annual standard deviation. The last column lists those statistics for the Topix index. The second panel repeats all statistics but now for the longer sample 1985–2021.

Mat. in years	1	2	3	4	5	10	15	Topix
Sample: 1996–2021								
Mean (Annual %)	0.13	0.34	0.63	1.01	1.41	3.34	4.67	5.13
St. Dev. (%)	0.25	0.53	0.92	1.32	1.69	3.14	4.54	24.08
Mean log (%)	0.13	0.34	0.63	0.99	1.39	3.24	4.48	2.49
Sample: 1985–2021								
Mean (%)	1.36	1.66	2.09	2.57	3.05	5.01	6.37	5.94
St. Dev. (%)	2.20	2.53	3.21	3.91	4.52	6.51	8.32	24.35
Mean log (%)	1.33	1.62	2.02	2.47	2.91	4.71	5.89	3.13

Table 12
Annual return differences between the Topix and constant maturity Japanese zero coupon bonds. The second row in the table lists the average difference between the returns on the Topix index ($\hat{\mu}^s$) and the annual returns on constant maturity zero coupon bonds ($\hat{\mu}_n^b$) for maturities ranging between 10 and 15 years using data between January 1996 and December 2021. The third row reports the t-statistic on the difference. The last row reports the annualized difference in the means of the annual log returns (instead of simple returns).

Duration in years	10	15
Sample: 1996–2021		
$\hat{\mu}^s - \hat{\mu}^b$ (% Annual)	1.79	0.46
t-stat on difference	0.36	0.09
Annual difference in mean log returns (% Annual)	−0.74	−1.98
Sample: 1986–2021		
$\hat{\mu}^s - \hat{\mu}^b$ (% Annual)	0.93	−0.43
t-stat on difference	0.22	−0.10
Annual difference in mean log returns	−1.58	−2.76

Table 12. Differences Topix minus long-maturity bonds show weakly positive differences in simple returns but negative differences in log-return means, reinforcing the message that the "equity premium" depends strongly on the benchmark and return metric.

6.4 Long sample world evidence and leverage-based duration matching (Figs. 12–15)

When full yield curves are unavailable over very long samples, the paper matches duration using leverage on a 10-year government bond instrument. Let $R^b_{t,10}$ be the return on the observed 10-year instrument, and let its duration be Dur_{10}. For a target duration Dur:

$$R^b_{t,Dur} = R^f_t + \frac{Dur}{Dur_{10}}\left(R^b_{t,10} - R^f_t\right). \tag{31}$$

Interpretation of Eq. (31). This is a duration-scaling approximation: for small yield moves, price changes scale roughly linearly with duration, so levering the 10-year bond's excess return constructs a higher-duration exposure.

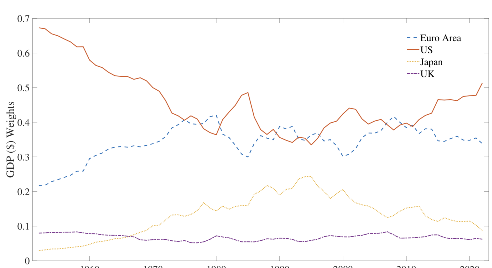

Fig. 12. GDP weights ($) World bank data. The graph plots the relative $ GDP weights of the Euro area, the US, Japan and the UK between 1952–2022.

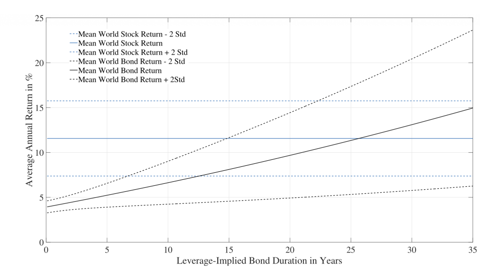

Fig. 13. Average world equity returns vs. World bond returns. The graph plots the GDP-weighted average world stock return (solid blue line) between 1953 and 2022 as well as the 95% confidence interval of the mean (dotted blue lines), and compares it to the GDP-weighted average world bond market portfolio return (solid black line) between 1953 and 2023 for varying leverage-implied durations (Dur) on the x-axis. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

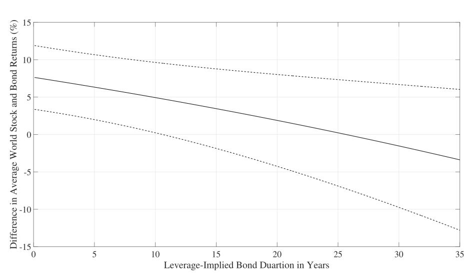

Fig. 14. Difference between average world equity returns and world bond returns. The graph plots the average difference between the GDP-weighted annual world stock return and the GDP-weighted annual world bond market portfolio return between 1953 and 2022 as well as the 95% confidence interval of the mean (dotted lines) for varying leverage-implied durations (Dur) on the x-axis.

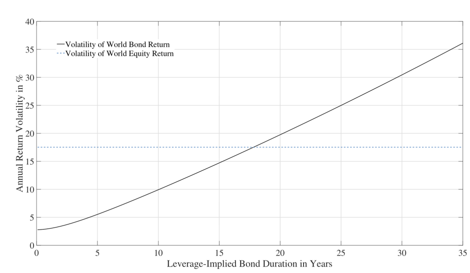

Fig. 15. Volatility of world equity returns vs. world bond returns. The graph plots the annual volatility of the GDP-weighted world stock return (dotted blue line) between 1953 and 2022 and compares it with the volatility of the GDP-weighted average world bond market portfolio return (solid black line) for varying leverage-implied durations (Dur). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Figs. 12–15. Using GDP weights to form world portfolios:

- **Fig. 12** shows the GDP weights (Euro area, US, Japan, UK) over time.
- **Fig. 13** plots mean world equity returns vs mean world bond returns as a function of leverage-implied duration. Mean estimates intersect at a long duration (around the mid-20s years in the paper's discussion).
- **Fig. 14** plots the mean difference (equity minus bond) vs duration: the conventional equity premium is the leftmost point, and the difference declines as duration matching increases.
- **Fig. 15** compares volatilities: bond volatility rises with duration and can match equity volatility at moderate-to-long durations.

6.5 US real bonds: additional evidence (Figs. 16–17)

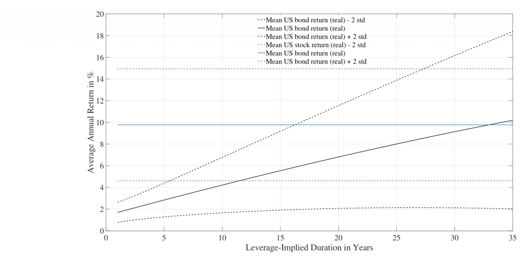

Fig. 16. Average US real equity returns vs. US real bond returns. The graph plots the average US real stock return (solid blue line) between 1983 and 2022 as well as the 95% confidence interval of the mean (dotted blue lines), and compares it to the average real bond market return (solid black line) for varying leverage-implied durations (Dur) on the *x*-axis. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

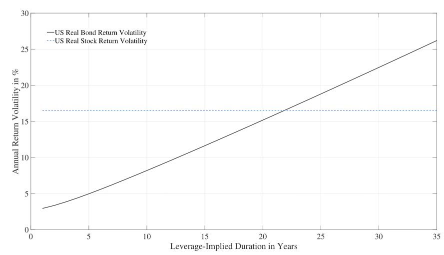

Fig. 17. Volatility of US real equity returns vs. US real bond returns. The graph plots the annual volatility of US real stock returns (solid blue line) between 1982 and 2022 and compares it with the volatility of US real bond returns (solid black line) for varying levels of leverage-implied durations (Dur). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Figs. 16–17 repeat the duration-matching logic for real bonds (model-implied real rates). They show that for sufficiently long duration, real bond returns can be comparable to real equity returns, and bond volatility increases with duration in ways similar to nominal-bond evidence.

7 Mechanisms and benchmarking (Fig. 18, Table 13, Eqs. 32–33)

7.1 Secular stagnation intuition via steady-state relations

A simple consumption-based steady-state link connects growth and interest rates. Assume CRRA utility:

$$U = \mathbb{E}_0 \left[\sum_{t=1}^{\infty} \beta^t \frac{C_t^{1-\gamma}}{1-\gamma} \right], \tag{32}$$

with lognormal consumption growth (mean g, variance σ_c^2). Then the steady-state interest rate satisfies:

$$y = -\ln(\beta) + \gamma g - \frac{1}{2}\gamma^2 \sigma_c^2. \tag{33}$$

Link to the paper's findings. Unexpected declines in long-run growth g can lower long rates y ("secular stagnation"). Long-duration bonds benefit mechanically from falling rates; equities may not, because lower growth also depresses future dividends, partially offsetting discount-rate gains.

7.2 Fig. 18: yield decomposition

Fig. 18 decomposes the 10-year yield into a risk-neutral yield and a term premium using the Adrian–Crump–Moench (ACM) framework. The paper's reading is that large time variation in

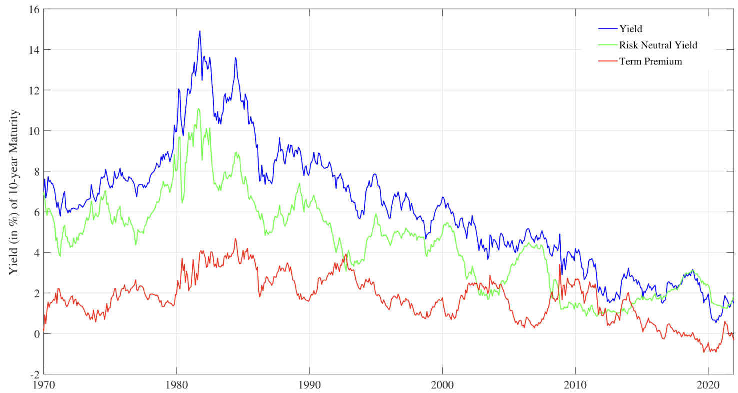

Fig. 18. Decomposition of 10-year US government yield. The figure plots the 10-year US government zero coupon bond yield and decomposes it into the term premium and the risk neutral yield as provided by Adrian et al. (2013).

Table 13
Percentiles of simulated annual returns on constant maturity 30-year zero coupon bonds. The second through sixth rows in the table list the percentiles of the moments of 10,000 simulated 50-year annual return samples for 30-year zero coupon bonds as implied by the ACM model. The second column reports percentiles of the estimated mean annual return (in %), the third column reports the percentiles of the estimated unconditional standard deviation of annual returns (in %) and the last column reports the percentiles of the Sharpe ratio.

Percentile	Starting at 1971			100-year burn-in		
	$\hat{\mu}^b$	St. dev.	Sharpe	$\hat{\mu}^b$	St. dev.	Sharpe
5	5.89	30.1	0.000	4.15	30.16	0.00
10	6.66	31.6	0.023	5.35	31.57	0.02
50	9.69	37.4	0.125	9.64	37.49	0.12
90	13.02	44.6	0.214	14.08	44.63	0.21
95	13.98	47.0	0.239	15.45	47.01	0.24
Data	13.39	30.8	0.271	13.39	30.8	0.271

yields does not necessarily imply a strong declining trend in term premia over the sample, so "declining bond risk premia" alone is not a sufficient explanation.

7.3 Table 13: model-based benchmark for long-bond moments

Table 13 compares realized long-bond return moments to simulations from an estimated affine term structure model (ACM). The key message is that standard stationary term-structure models struggle to reproduce the combination of high mean returns and high volatility observed for very long duration bonds in the data.

8 Robustness: payout measurement and tradable bond funds (Table 14, Figs. 19–20)

8.1 Table 14: dividend yield vs total net payout yield

Table 14 reports average dividend yield and total net payout yield (dividends + repurchases − issuance). In the data used, net issuance dominates, making total net payout yield smaller (even negative in the recent subsample). Thus, using net payout would *increase* implied duration (lower yield), which would strengthen—rather than weaken— the duration-matching logic.

Table 14
Average annual dividend yield and total net payout yield in the U.S. The table reports the value weighted average annual dividend yield as well as the average net payout yield (the dividend yield plus the stock repurchase yield minus the stock issuance yield) for all U.S. stocks in the CRSP database as provided by Boudoukh et al. (2007) over the 1970–2020 sample, as well as over the most recent subsample (1996–2020) in percent.

	Dividend yield	Total net payout yield
1970–2020	2.79%	1.09%
1996–2020	1.90%	−0.09%

8.2 Figs. 19–20: validating yield-curve-implied bond returns

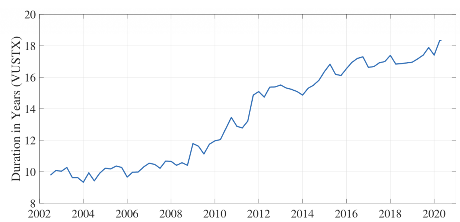

Fig. 19. Vanguard long-term government bond index fund. The figure plots the effective duration of the Vanguard Long-Term Government Bond Index Fund (VUSTX).

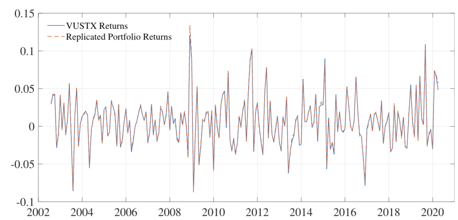

Fig. 20. Vanguard long-term government bond returns (VUSTX) vs. Replicated returns. The graph plots the monthly returns to investors on Vanguard's long-term government bond index fund (VUSTX) and plots it against replicated returns on duration-matched portfolios based on the zero curves provided by Gurkaynak et al. (2006).

Figs. 19–20 use the Vanguard long-term government bond index fund (VUSTX):

- Fig. 19 plots the fund's effective duration over time.
- Fig. 20 compares actual monthly VUSTX returns to replicated duration-matched returns constructed from the GSW zero curves.

The close match supports the quality of the yield-curve-based bond return construction used earlier in the paper.

9 Conclusion

9.1 Summary

This paper concludes that many standard statements about (i) stock-market "outperformance" and (ii) stock-market "excess volatility" are partly artifacts of a **duration mismatch in benchmarking**. The conventional benchmark for equity performance is a short-duration risk-free asset (e.g., T-bills). But aggregate equity is a **long-duration** claim on dividends. Once stock returns are evaluated relative to **duration-matched government-bond counterfactuals**—fixed-income portfolios constructed to match the duration profile of aggregate dividends (via dividend strips)—the historical picture changes sharply.

The central conclusions are:

(i) **Equities do not robustly outperform duration-matched government bonds.** Over long samples (e.g., roughly 1971–2021, and in some readings even longer), the realized excess performance of aggregate stock indices over these duration-matched bond counterfactuals is often *small* and can be economically modest even when the classic stock–T-bill premium is large.

(ii) **Long-duration government bonds can be as volatile as, or more volatile than, stocks.** When bonds are selected to have durations comparable to equity (dividend) duration, their realized volatility can match or exceed stock-index volatility. Hence, the idea that stocks are uniquely "excessively volatile" is substantially weakened under a duration-consistent benchmark.

(iii) **These findings speak directly to the equity premium and excess-volatility (bubble) puzzles.** A nontrivial portion of what looks like a large "equity premium" against T-bills can be reinterpreted as compensation associated with holding *long-duration* assets during regimes of large interest-rate movements, especially long secular declines in yields.

(iv) **The results are not explained away by mechanical accounting adjustments.** The paper argues the conclusions are not eliminated by net repurchases (buybacks) or by inflation adjustments, reinforcing the benchmark mismatch interpretation.

(v) **Ex post evidence does not pin down ex ante expectations.** Observing little realized compensation for long-duration dividend risk *ex post* does not imply investors did not expect compensation *ex ante*. The gap between expected and realized premia depends on the evolution of long-run growth expectations, discount-rate expectations, and risk premia over time.

9.2 Why duration-matching changes the conclusions

1) The benchmarking problem is a duration mismatch. A stock index is a claim on a stream of dividends far into the future (high duration). A T-bill is a near-instantaneous claim (low duration). Comparing equities to T-bills therefore mixes two conceptually different sources of return:

- compensation for **equity cash-flow risk** and equity risk premia, and
- compensation for **long-duration exposure** (term risk and sensitivity to discount-rate movements).

2) Why the equity premium can look large against bills but small against long bonds. If the macro environment features long-run declines in interest rates (a "secular" fall in yields), *all long-duration assets* mechanically earn high returns because discount rates fall. A duration-matched long-bond portfolio will capture much of this gain. Therefore, the residual return difference between stocks and duration-matched bonds (a "duration-adjusted equity premium") can be much smaller than the classic stock–bill premium.

3) Why "excess volatility" can disappear under a duration-consistent benchmark. Volatility comparisons are also benchmark-dependent. Long-duration bonds are highly sensitive to changes in yields:

$$\text{Price sensitivity} \;\approx\; -D \cdot \Delta y,$$

so when duration D is large (especially in low-rate environments where duration can be very high), bond price volatility can be substantial. Once bond duration is made comparable to equity duration, bond volatility becomes a relevant counterfactual and can be similar to, or exceed, equity volatility. Thus, stocks may not be exceptionally volatile relative to *other assets with comparable duration exposure.*

4) A mechanism lens: secular stagnation as an example. One plausible macro explanation highlighted by the paper is that long-run declines in expected growth (or increases in desired saving) reduce long rates (a "secular stagnation" type force). Falling rates boost long-duration bond prices directly; for equities, the same force can simultaneously lower expected dividend growth, offsetting some of the discount-rate benefit. This naturally generates periods where bonds keep up with (or outperform) equities on a duration-matched basis.

9.3 Takeaway

If equity is benchmarked against fixed-income portfolios that match the duration of aggregate dividends, then historical stock "outperformance" and stock "excess volatility" look much smaller: long-duration government bonds have delivered similar realized performance and comparable (or higher) volatility, so duration mismatch is central to interpreting the equity premium and bubble narratives.